

Task 7: Business Insights & Recommendations

Based on the Exploratory Data Analysis (EDA), data visualizations, and Power BI dashboard, several important business insights were identified from the Supermarket Sales dataset. These insights help in understanding sales performance, product-line performance, city-wise performance, monthly trends, customer behavior, and payment preferences.

1. Business Insights

Insight 1: Food and Beverages is the Highest-Selling Product Line

The Food and Beverages product line has the highest total sales among all product lines. This indicates strong customer demand for food and beverage products. On the other hand, Health and Beauty has the lowest total sales among the product lines.

Business meaning: The supermarket should continue maintaining sufficient stock of Food and Beverages while exploring ways to improve the performance of Health and Beauty products.

Insight 2: Naypyitaw is the Highest-Performing City

Among the three cities, Naypyitaw records the highest total sales, while Mandalay has the lowest total sales, although the difference between the cities is relatively small.

Business meaning: Naypyitaw shows slightly stronger sales performance and can be studied to understand which products and customer behaviors are contributing to its performance.

Insight 3: Sales Vary Across the Three Months

The monthly sales analysis shows that January has the highest total sales, while February has the lowest total sales. Sales increase again in March after the decline in February.

Business meaning: The variation in monthly sales indicates that customer demand changes over time. The supermarket can use monthly sales patterns to plan inventory and promotional activities.

Insight 4: E-wallet is the Most Used Payment Method

The Power BI dashboard shows that E-wallet is the most frequently used payment method, followed closely by Cash, while Credit Card has the lowest number of transactions.

Business meaning: Customers are increasingly using digital payment methods. Supporting convenient digital payment options can help provide a better customer experience.

Insight 5: Sales Performance Differs Across Product Lines

The sales comparison shows that the product lines have different levels of performance. Food and Beverages, Sports and Travel, and Electronic Accessories are among the stronger-performing product lines, while Health and Beauty records comparatively lower sales.

Business meaning: Product-line performance can help the supermarket decide where to focus inventory, promotions, and marketing efforts.

Insight 6: Sales are Strongly Connected with Quantity Sold

The analysis of numerical variables shows a strong positive relationship between Quantity and Total Sales. When more units are purchased, the total transaction value generally increases.

Business meaning: Increasing the number of items purchased per transaction through suitable offers and product combinations can contribute to higher sales.

Insight 7: Customer Types Have Similar Representation

The dataset contains both Member and Normal customers in almost equal proportions. This indicates that both customer groups contribute significantly to the supermarket's transactions.

Business meaning: The supermarket should maintain good service for both groups while using membership benefits and targeted offers to encourage repeat purchases and customer loyalty.

2. Important Trends and Patterns

- Food and Beverages is the highest-performing product line based on total sales.
- Health and Beauty is the lowest-performing product line based on total sales.
- Naypyitaw has the highest total sales among the three cities, while Mandalay has the lowest.
- January records the highest monthly sales, while February records the lowest.
- E-wallet is the most commonly used payment method.
- Sales performance varies across product lines and cities, showing opportunities for targeted business strategies.

- Higher quantities purchased generally contribute to higher total transaction values.

2. High and Low Performing Areas

➤ High-Performing Areas

- Product Line: Food and Beverages
- City: Naypyitaw
- Month: January
- Payment Method: E-wallet

➤ Low-Performing Areas

- Product Line: Health and Beauty
- City: Mandalay
- Month: February
- Payment Method: Credit Card has the lowest transaction usage

4. Data-Driven Business Recommendations

Recommendation 1: Promote Low-Performing Product Lines

The supermarket should introduce targeted discounts, combo offers, and promotional campaigns for the Health and Beauty product line to improve its sales performance. Customer purchase patterns can be analyzed further to create suitable product bundles.

Recommendation 2: Improve Performance in Lower-Selling Cities

The supermarket should analyze the reasons behind the comparatively lower sales in Mandalay and introduce location-specific promotions, product offers, and marketing campaigns to increase customer engagement and sales.

Recommendation 3: Plan Inventory According to Monthly Demand

Since sales vary across months, the supermarket should use historical sales trends for inventory planning and promotional scheduling. Higher-demand periods should have sufficient stock, while promotional activities can be used during lower-sales periods such as February.

Recommendation 4: Encourage Digital Payment Usage

Since E-wallet is the most frequently used payment method, the supermarket can continue supporting digital payments and introduce suitable digital-payment offers or loyalty benefits to improve customer convenience and engagement.

Task 7 Conclusion

The analysis of the Supermarket Sales dataset identified important patterns in product-line sales, city-wise performance, monthly sales, payment methods, and customer behavior. Food and Beverages and Naypyitaw were identified as strong-performing areas, while Health and Beauty and Mandalay showed comparatively lower performance. Monthly sales also showed variation, with January performing better than February.

Based on these findings, the supermarket can improve business performance by focusing on low-performing product lines and locations, planning inventory according to demand patterns, and supporting preferred payment methods. These data-driven insights and recommendations can help management make better decisions related to sales, marketing, inventory, and customer engagement.